

Optimizing Acconeer A121 Pulsed-Coherent Radar Distance Readings

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1. Pulsed-Coherent Radar

1.1 Operation

Acconeer's A121 radar module sends pulsed 60.5 GHz radar signals from a single TX and a single RX antenna. These ~5mm wavelength signals can then be used to resolve object distances by examining the time of flight, amplitude, and phase of the reflected waves. The A121 radar is considered a *pulsed* radar sensor because it sends configurable, short TX signals in pulses to be reflected off of a target. The duration and amplitude of these signals determines the precision of measurement results. The A121 radar is considered *coherent* because it preserves the phase of reflected signals, which is critical for extremely precise distance measurements.

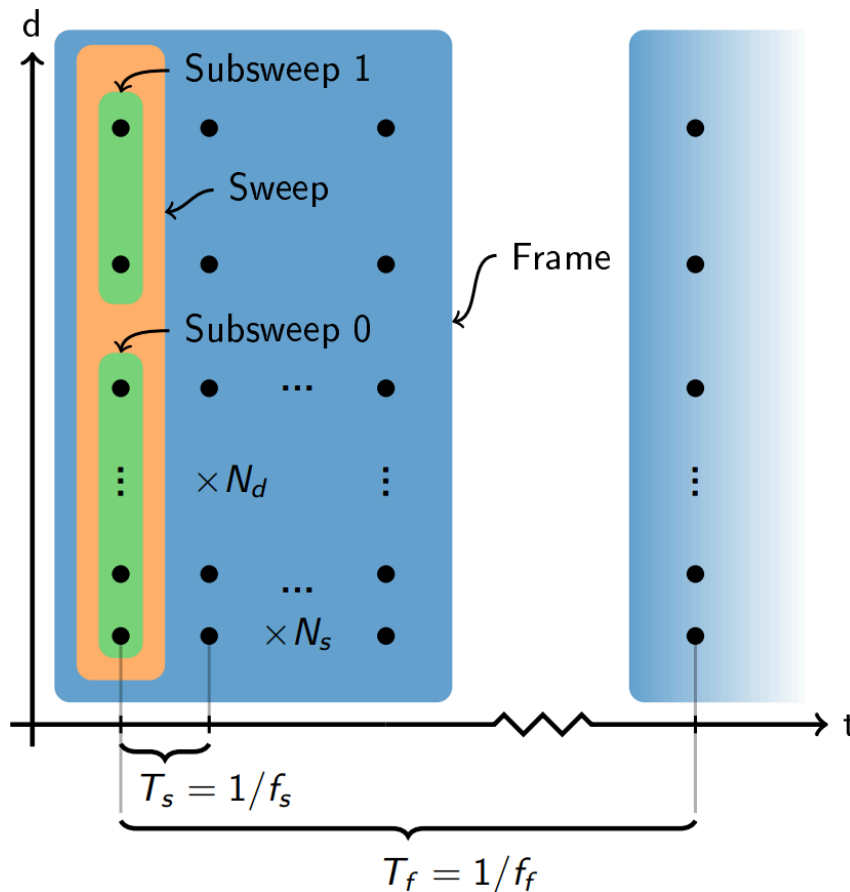


Figure 1.1.1 An illustration of Acconeer's measurement hierarchy. Each frame contains several subsweeps, which contain a series of 60.5 GHz pulses corresponding to each distance bin.

When measuring distances, the Acconeer RSS software on the A121 module takes a distance measurement **frame**. A frame consists of several **sweeps**, which are a series of pulses that are

sent at times that correspond to different range **bins** (*Figure 1.1.1*). A range bin is a discrete segment along the sensor's line of sight that has a corresponding IQ value of a reflected signal. Once a frame has been taken, the RSS software can resolve objects from the IQ data in the frame. This is accomplished by examining the amplitude of the averaged IQ values across sweeps, and then comparing them to either a predetermined or dynamic strength threshold. Amplitudes above the threshold are then resolved into objects, with a distance corresponding to the bin of the amplitudes. Notably, documentation states that the resolution of this strategy is primarily dependent on SNR, granted that the desired resolution is greater than half the wavelength, meaning that any resolution under 2.5mm is hard to obtain (*Figure 1.1.2*). It is important to consider that much more precise measurements are possible within each bin by examining the phase of the reflected wave, which is something that acconeer's distance detector software does not do.

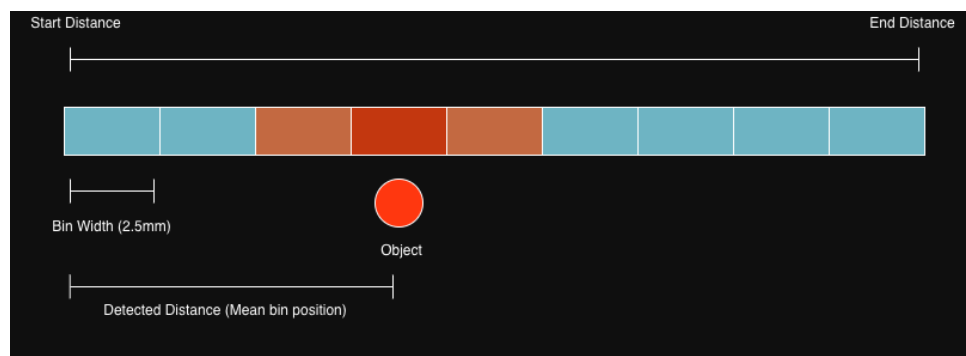


Figure 1.1.2 A visualization of an A121 sensor resolving object distances using bins. The sensor sends out TX pulses timed to correspond to each bin. A pulse registers as above threshold in several bins, which can then be resolved by examining the amplitude in each bin. Phase in the I2C application is only used to verify that detected pulses are from the same object.

1.2 Strength Measurement

Acconeer's I2C applications for the XM125 module, which breaks out the A121 distance sensors with an STM32L4 chip and provides I2C and UART interfaces, gives both distance and strength readings for each distance measurement command. The strength reading does not correspond to the raw amplitude of the received pulses, and instead is a value dependent on the distance reading, phase, and amplitude of the received pulses that is meant as a way to characterize objects. This means that as an object moves further away from the sensor, strength readings from subsequent distance measurements will remain relatively constant, as the firmware compensates for the decreasing pulse amplitude.

Strength readings are highly angle-dependent. Even small changes in the angle at which each A121 pulse hits the object of interest can cause large differences in strength that disrupt object characterization. This means that in deployment, a swaying sensor, or borehole walls that grow in ways that change the angle of the reflected pulse, will make using strength readings to identify valid distance measurements not feasible. In testing, in a static environment with

consistent angles, strength readings are useful to reject distance measurements that come from second-order reflections and pulses penetrating into dielectric materials.

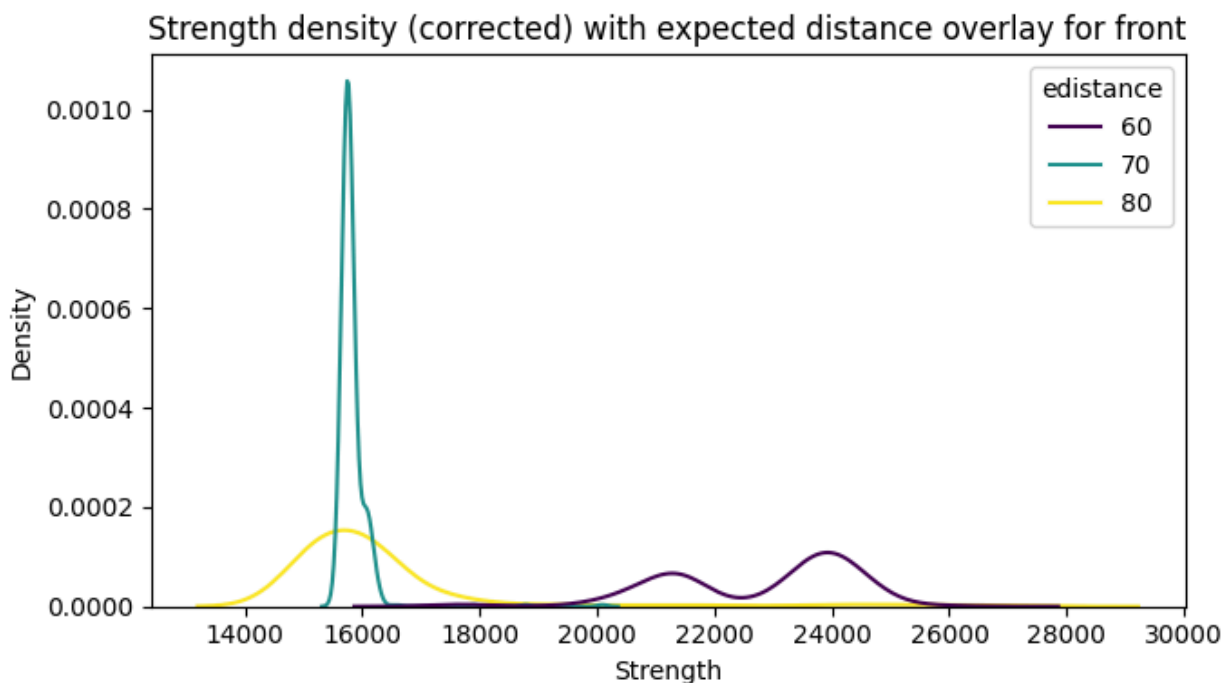


Figure 1.2.1 A distribution of strength readings for a dielectric PVC tube at different distances. At each distance, 200 strength readings were taken while the tube was vibrated. The curved outer surface of the tube, plus the small vibrations, causes changes in angle between each observation that disturb strength readings, which should have a sharp theoretical distribution. It is apparent that the tube most likely has an adjusted strength value of 16000 (which translates to -16 dB), but noisy readings from slight changes in angle cause spikes at -21 dB and -24 dB.

1.3 Sensor Recalibration

The XM125 using the I2C application can send a recalibration notice when it detects a large change in temperature or a drift in its measurements. Calibration of the sensor involves two steps, first doing an electronic loopback test on-sensor to verify TX and RX are timed correctly, and then sending out test pulses into the environment to better filter out ambient noise. This recalibration can “bake” objects of interest into the background environment, and result in subsequent distance measurements not detecting the object until it moves.

While recalibration is extremely important, it is also dangerous to recalibrate in an environment where the object of interest may only move 5 mm every year. A change in temperature may cause the sensor to automatically recalibrate, baking the object of interest into the environment and causing subsequent distance measurements to not trigger. It is important to manually and intentionally control sensor calibration to avoid this scenario. During sensor testing, the sensor calibrates on wakeup, and before each wakeup the testing object of interest was removed from the sensor FOV in order for it to not be accounted for in the calibration.

2. Configuring Parameters

2.1 Configuring for Precise Measurements

Acconeer provides several configuration options for the A121, which must be tuned in order to obtain precise and accurate distance measurements. Generally, a balance must be struck between the computational load of each measurement and the small processing power and memory of the STM32L4C chip that Acconeer uses to control the A121 on the XM125 module. “Overloading” the configurations will result in high power draw, slower measurements, and potential stalling of the module.

Step size, or the length of the bins in a sweep, must be minimized at 2.5mm for the most accurate measurement. Lowering step size increases the amount of pulses per sweep; halving the step size doubles the amount of pulses sent out, doubling the computation needed.

Measuring range, or setting the start and end bins for a sweep, must be minimized in order to preserve processing power. A larger measuring range results in more steps/bins and thus more pulses.

Profile, or the length of each pulse, must also be minimized. Smaller pulses provide higher distance resolutions, and larger pulses have potential to overlap between bins. For absolute distance accuracy, smaller profiles must be used. However, smaller profiles result in worse SNR as they are shorter and less prominent against background noise, and in measurement-intensive applications they are also less power efficient, as more sweeps are required to achieve SNR targets. A balance must be struck between profile (pulse length) and maximizing signal quality.

Tuning **signal quality** adjusts the target SNR ratio on the A121, and must be maximized. This means on average, a higher signal quality parameter (in dB), will result in higher amplitude pulses as the device attempts to maintain a higher SNR against background noise. Higher amplitude pulses results in more power consumption. This tuning is achieved internally by adjusting the high water automatic average sample (HWAAS).

The **threshold method** determines how the A121 chooses to resolve distances against the background noise. Advanced methods use an adaptive threshold based on previous measurements, and require more computation. Fixed strength and amplitude thresholds accept any signals above a fixed value.

2.2 Optimal Configuration for Static Measurements

To achieve the highest precision SNR must be maximized. Choosing the smallest step length (2.5mm), highest desirable SNR ratio (35 dB), and correct profile are all important to maximize SNR. It is not necessarily obvious which profile provides the best SNR, as lower profiles provide shorter, more resolvable pulses, but larger profiles are easier to detect against background noise.

To determine the optimal parameter configuration for the A121, distance measurements were conducted that resolved the distance of a dielectric HPDE rod at 5 mm and 2 mm intervals for different profile configurations in order to determine the optimal profile. For each trial, bin step length was held at 2.5 mm, close range leakage cancellation was enabled, start distance was at 90 mm, end distance was at 150 mm, and SNR was at 35 dB (*Figure 2.1, Figure 2.2, Table 2.1, Table 2.2*). At each distance, 60 measurements were taken, outliers greater than 10 mm from expected value were rejected, and then remaining values were averaged.

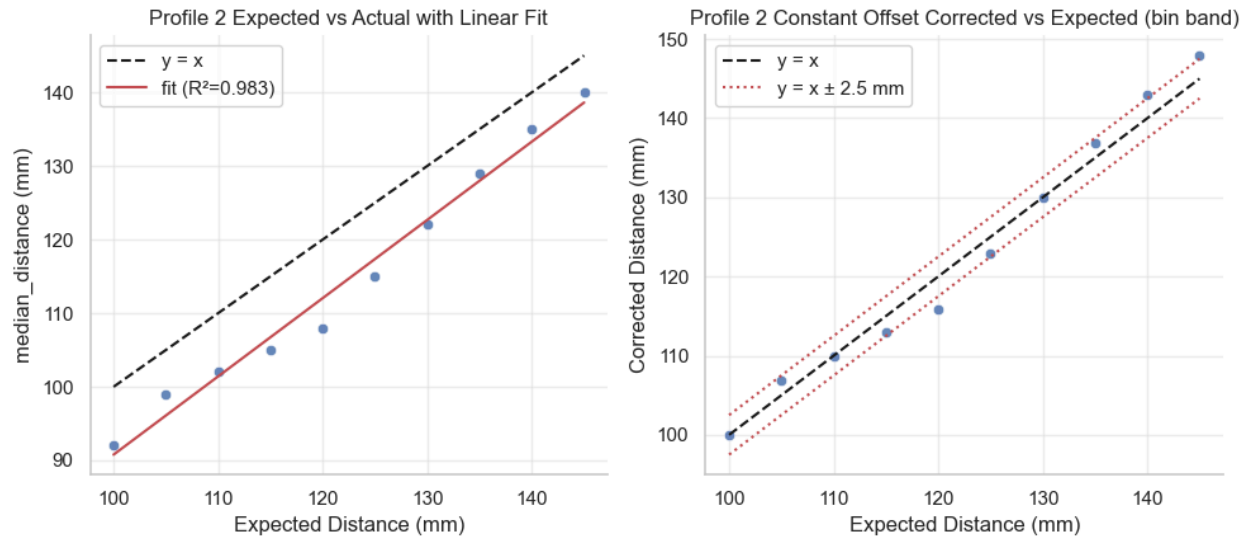


Figure 2.2.1 Plot A shows expected distance vs measured distance at 5mm increments from 100mm to 145mm for profile 2 with a linear fit ($m = 1.06$). Using a constant offset model, the corrected readings are displayed in plot B, where a 2.5mm resolution band is displayed. Almost all measurements are within the band, which is consistent with Acconeer's A121 documentation. .

Profile	Correction (mm)	Measurement Error (mm)	Std Error (mm)	% Within 2.5 mm Band	N
1	-8.342	-1.55e-15	1.368	88	421
2	-7.995	-4.17e-15	2.099	78	463
3	-7.913	-1.16e-15	3.142	68	487

Figure 2.2.2 Lists correction offset, error in measurements, standard deviation from expected measurement with offset applied, percent of measurements within 2.5mm resolution with offset applied, and number of measurements for trials from 100mm - 145mm with 5mm step between trials.

We see that profile 1 and 2 are comparable, but profile 2 appears to be slightly more stable, especially when considering strength-rejected data (where internal/non-strength matching measurements are not included in the analysis), while profile 1 has slope closer to 1. Notably, a cyclic pattern can be seen in all of the trials, where measurements tend to alternate between over and underestimates (when a constant offset is applied to account for systematic bias). All measurements are almost all within the $\pm 2.5\text{mm}$ resolution range that the documentation suggests. Deployment of the sensor will most likely involve using profile 1, power constraint dependent.

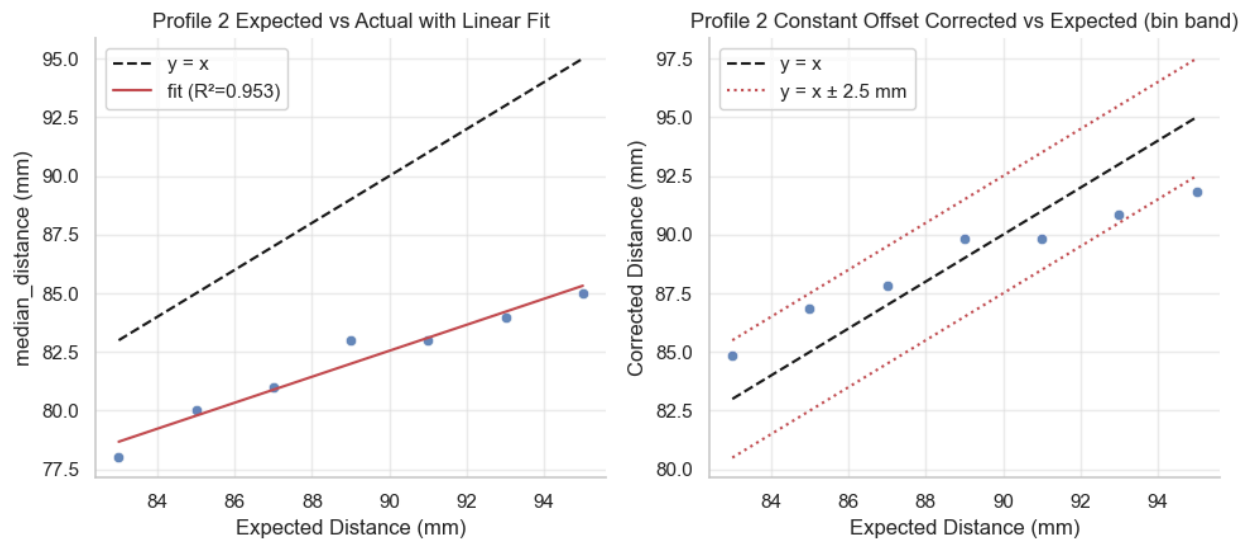


Figure 2.2.3 Shows expected distance vs measured distance at 5mm increments from 100mm to 145mm for profile 2 with a linear fit ($m = 0.586$), and a constant offset model with a 2.5mm resolution band.

Profile	Correction (mm)	Measurement Error (mm)	Std Error (mm)	% Within 2.5 mm Band	N
1	-7.063	-4.57e-15	2.107	66	348
2	-6.841	-1.41e-15	1.77	91	321

Figure 2.2.4 Lists correction offset, error in measurements, standard deviation from expected measurement with offset applied, percent of measurements within 2.5mm resolution with offset applied, and number of measurements for trials from 100mm - 145mm with 5mm step between trials.

Using profile 2 and the max-precision settings, we now take measurements from 85mm-95mm at a set measuring range of 75-100mm at 2mm increments. We see that these measurements are poor. Notably, almost all of these measurements are within the 2.5mm resolution envelope,

and the cyclic under/overestimation is again apparent, which is what I believe accounts for the 0.59 best fit slope.

Using these settings on the I2C application, within this testing setup, it seems that any resolution approaching 2.5mm is not attainable. The device *can* detect on the order of 1-2mm changes, but each measurement is subject to the resolution limit of ~2.5mm. I've found that using 4-5mm increments in testing is the threshold for stable, linear models to reliably resolve change in distance. For the purpose of detecting shifts in borehole diameter, I am confident that the sensor will detect changes, but I am not confident in the sensor's ability to accurately model changes that are on the order of 1-2mm.

Upon contact, Acconeer stated that in ideal conditions, standard deviation of distance sensor application measurements is 2.67 mm. They prefaced this number by saying that changes in temperature, frost forming on the sensor, multipath reflections, and the curvature of tunnel walls will likely worsen the standard deviation, which is consistent with my benchmarking. Acconeer suggested that I track the phase over a single distance bin (the phase rotates 360 deg over 2.5mm) at a fixed distance to measure the ice progression, and warned that this would still be vulnerable to temperature and multipath reflections. This is in line with the idea of a hybrid approach, involving tracking the phase of the measurements by developing a custom I2C endpoint for the STM32 firmware, and also using the standard I2C distance application to estimate the absolute distance.

2.3 Differential Measuring

For the purpose of borehole diameter logging, two XM125 sensors will be mounted on opposite sides of the PCB. Change in tunnel diameter will be measured as the difference between the distance readings of the two sensors. This method, while allowing for disturbance rejection, is vulnerable to reflections and multipath within the circular cavity the sensors are suspended in.

Differential Measurement testing was done by suspending the sensing module inside a 20cm diameter HPDE tube, and then measuring with both sensors at 5mm increments while using profile 2 and the previously mentioned high accuracy settings. Changes in distance were then calculated by taking the difference between the distance read on the front sensor and the distance read on the back sensor. For each distance, measurements were taken 100 times.

Figure 2.3.1 displays the differences between the distances measured by the front and back sensors as the HPDE tube was moved in increments of 5mm while the sensor PCB remained fixed. The expected trend line is overlaid on the results, where there is a noticeable difference. The inaccuracy of the differential measurements is likely due to the curvature of the tube and multipath reflections in the tunnel. While the movement of the tube is clearly visible, there is a clear discrepancy between the data and the model.

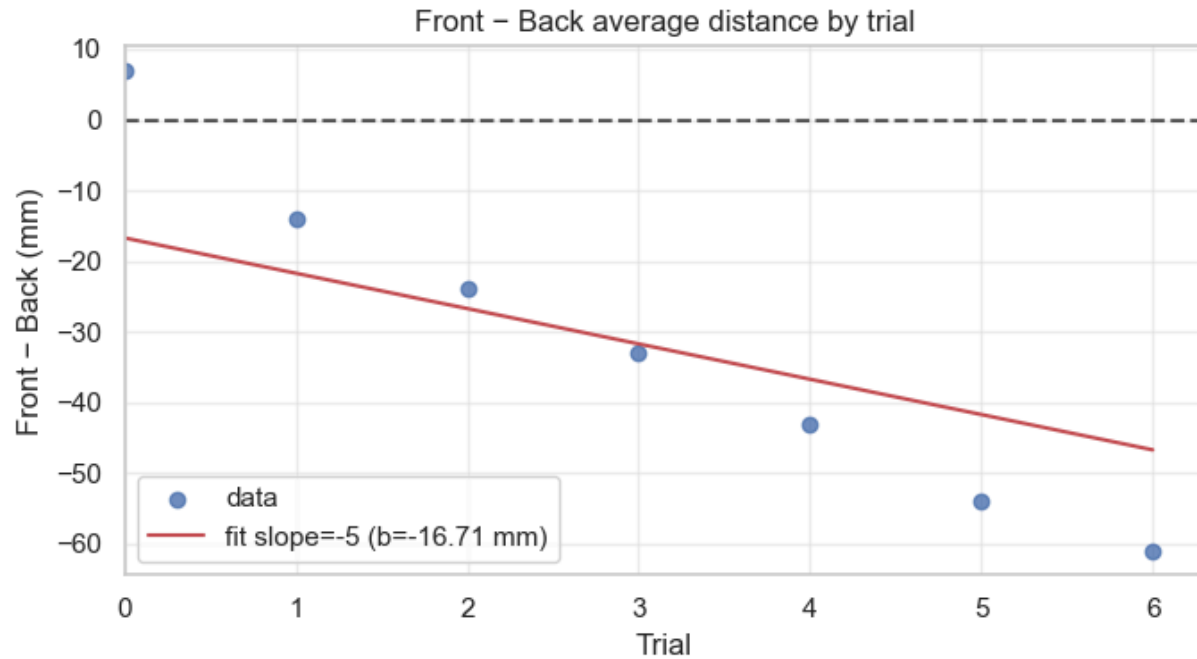
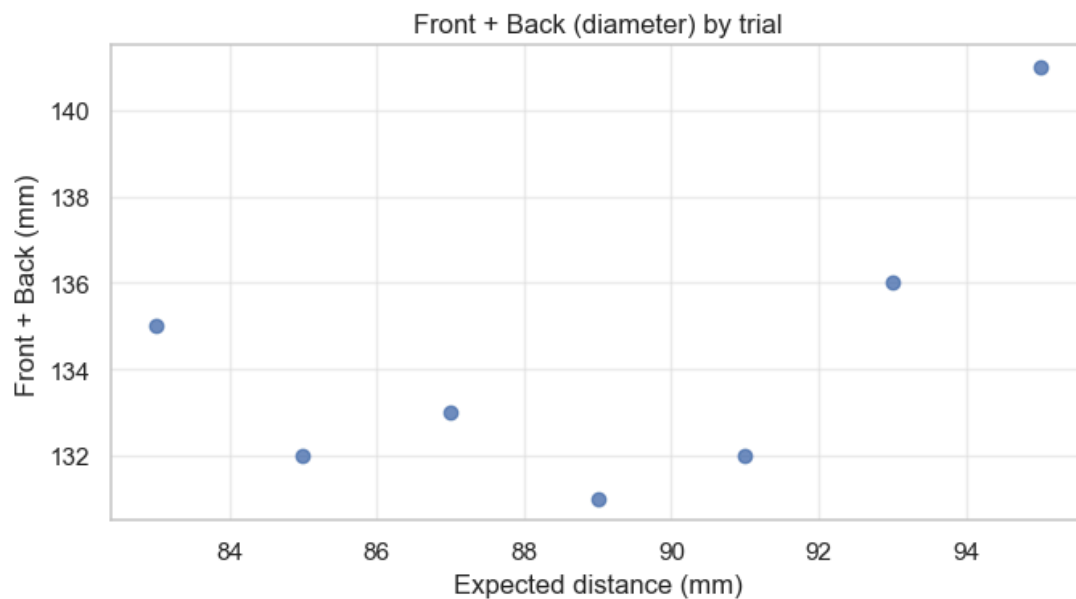


Figure 2.3.1

Figure 2.3.2 shows the calculated diameter of the tube for each trial, which is simply the sum of the front and back distance measurements. The actual tube diameter was 135mm. The measurements fluctuate around this diameter, hinting at large sources of error that need to be investigated further. For the purpose of borehole diameter tracking, it is crucial that these measurements are accurate.



3. Comments

3.1 Conclusion

The XM125 I2C user-level applications were found to provide distance measurements with at least 2.5mm of error, which is in line with Acconeer's stated 2.67mm error. While this resolution is still high enough for detecting large changes in the ice in RNO-G boreholes, the curved nature of the holes, multipath reflections from the tunnel, and temperature fluctuations will likely lead to errors higher than 2.5mm.

In order to effectively log borehole diameter, an approach that uses both the raw IQ data and processed distance results from the XM125 will be needed. To expose the raw IQ data to a master microcontroller, custom I2C endpoints will be developed on top of Acconeer's I2C firmware. Using this approach, sub-millimeter level relative changes in tunnel diameter will be able to be detected based on the shifted phase of reflected pulses. This approach is still vulnerable to the curved surface and multipath of being in a tunnel, as well as temperature fluctuations.

3.2 Reference Documents

- 1) A121 Pulsed Coherent Radar Sensor
https://www.mouser.com/pdfDocs/A121datasheet1.pdf?srsId=AfmBOorgrk4ic_ecZ16UF3ncci-5iebLkRNnhBi9N8Fv7v_R3tUBy6
- 2) Acconeer XM125 Parameter Configuration Guide
https://docs.acconeer.com/en/latest/radar_data_and_control/a121/how_to_configure.html
- 3) XM125 Datasheet
https://www.mouser.com/pdfDocs/XM125_datasheet-3133248.pdf?srsId=AfmBOoroZYFu-6pvaE4vxuurckhFaR3dDneTYPFu6H_JpP7bCPhuJ7h
- 4) XM125 Software User Guide
- 5) XM125 Distance Detector User Guide